

Multimodal RAG Studio · Sample Document

This document describes a transformer-based architecture for multimodal question answering. The system combines text and visual evidence to produce grounded answers.

The architecture consists of an encoder, an attention mechanism, and a decoder. The encoder processes the input, the attention layer computes relevance weights, and the decoder generates output.

2. System Architecture

The figure below shows the overall system design.

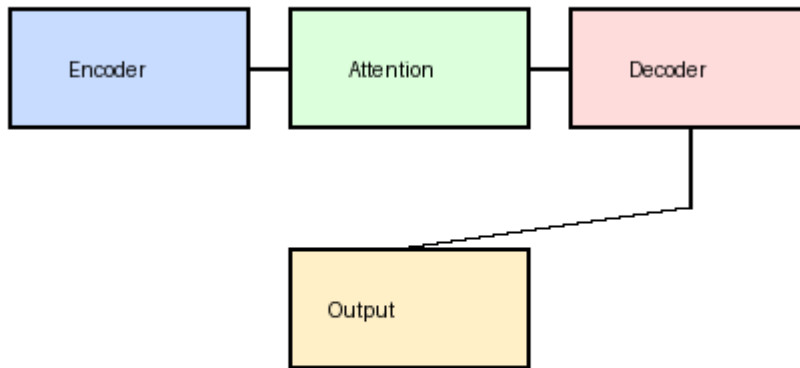


Figure 1: The overall system architecture showing encoder, attention, and decoder components.

3. Attention Mechanism

The attention layer computes a weighted sum of values based on query-key similarity. This allows the model to focus on the most relevant parts of the input when producing each output token.

Performance improved significantly after adding multi-head attention.

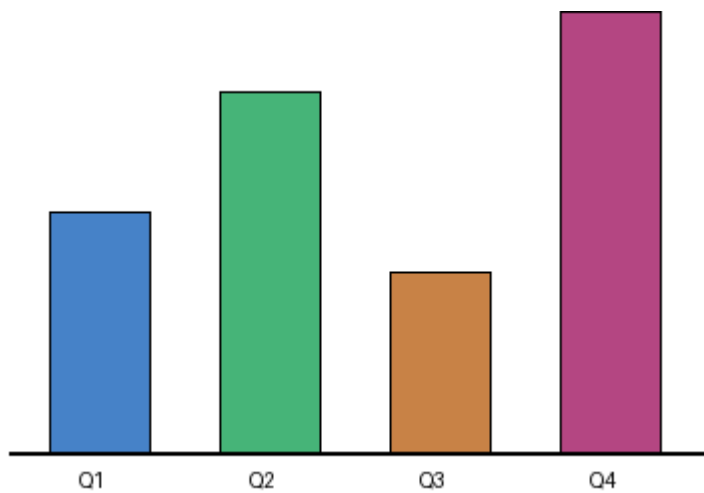


Figure 2: Quarterly accuracy after introducing multi-head attention.

4. Results

Table 1: Model comparison on the test set.

Model	Accuracy	Latency
Transformer	92.3%	45ms
LSTM	85.1%	30ms
CNN	78.4%	20ms

The transformer achieves the highest accuracy, though with higher latency due to the attention computation.